

Embedded Systems Development

Task 3.1P Create Trigger/Notification based on Sensor Data

For real-world projects, there would be number of times that you are required to perform an action given certain conditions are true. For example, generate a SMS or an email when a payment is received or when the temperature of the server room goes beyond the accepted level. There are different platforms which allows you to do this using Webhooks. For example, **IF This Then That (IFTTT)** is a web service that provides a platform for services running on various devices and application to be monitored and triggered, hence enabling you to easily integrate multiple services.

In this task, you will build up on your knowledge on collecting data from sensors and using specific communication protocols such as I2C to build a simple embedded system that will generate a trigger and a notification based on certain conditions.

The task can be completed using both IFTTT platform and Node-RED. However, IFTTT is a paid platform and will only allow you to have access free for a week.

Hardware Required

- Arduino 33 IoT
- Light sensor
- Breadboard and jumper wires
- Resistors

Software Required

- For IFTTT
 - IFTTT registered account or
- For Node-RED
 - Node-RED installed on a Raspberry Pi or PC
 - MQTT Broker (e.g., Mosquitto or HiveMQ Cloud)
 - Email or Telegram notification service (configured in Node-RED)

Pre-requisites

1. Complete all previous Arduino tasks.
2. For IFTTT
 - a. Research on the use of IFTTT. Arduino provides a very good documentation in [Sending an email with IFTTT](#)
 - b. Set up an IFTTT account using your student email address [IFTTT Official Site](#)

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Task Objective

This task requires you to build a trigger and a notification when certain light level conditions are met using either IFTTT or Node-RED/MQTT.

Problem statement:

Linda has just bought a beautiful terrarium for the house. she was advised to keep the terrarium in sunlight for an average of 2 hours a day. You aim to build an embedded device that uses the Arduino device to measure the amount of sunlight exposure of her terrarium. You can use either IFTTT or Node-RED services to generate notifications when the sunlight hits your terrarium and another notification when it stops. The sunlight information is to be provided by your Arduino device. You can use any notification mechanism that is supported by IFTTT and Node-RED-MQTT (mobile notification, email etc).

Steps:

1. Design your embedded solution integrating hardware and a trigger/notification mechanism built using either IFTTT or Node-Red.
2. Build the hardware set up using the Arduino Nano 33 IoT and the BH1750 light sensor
3. In the Arduino IDE, install the sensor library for BH1750 sensor. Run the example file provided in file->examples->BH1750->BH1750test to verify your sensor is collecting the data accurately.
4. **For those who are using IFTTT:** Follow the instructions provided in [Sending an email with IFTTT](#) to integrate IFTTT to send notifications
5. **For those who are using Node-RED:** Follow the instructions provided in [Sending email with Node-RED](#) to integrate Node-RED and MQTT to send notofications.
6. Test your embedded solution.

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Task Submission Details

1. Describe how your system works and clearly outline the following:
 - a) A block diagram of the entire solution including the hardware and the trigger/notification platform you decide to use.
 - b) How the trigger mechanism is handled in your code (you will need to include a screenshot of the relevant code and explain this)
 - c) How the notification mechanism is handled in your code (you will need to include a screenshot of the relevant code and explain this)
2. In less than two paragraphs, describe how you would test the system you have built to see if it is working as intended?
3. Create a folder named Task3.1Trigger on your 210/730 GitHub repository. Upload your code to the repository. Include the link to your repository there.
4. **For SIT730 students only**
Discuss if IFTTT can be used to implement communication between two Arduino Nano 33 IoT devices.

Important:

- Please note explanations, diagrams, or code directly taken from GenAI is NOT accepted in this unit. The tasks submitted should be your own work in accordance with the academic integrity policies of Deakin. Task submissions that do not adhere to [Deakin Academic Integrity Guidelines](#) will be reported to the Academic Integrity Team for further action. Please refer to [Ethical and Evaluative Use of GenAI](#) for more information. If you have used GenAI in certain parts of your task and not sure if that is allowed, please talk to the unit chair.
- If you need any clarifications on the task requirements or require help to complete the tasks, please contact your teaching team.
- Anytime you submit a task to OnTrack, it is a good practice to check the status of any existing tasks, and the future tasks you are expected to complete. If you have got feedback on previous tasks, you may need to fix and resubmit some of your work. You want to check out why, so that you can learn from this and make it faster and easier to accomplish later work to the required standard.